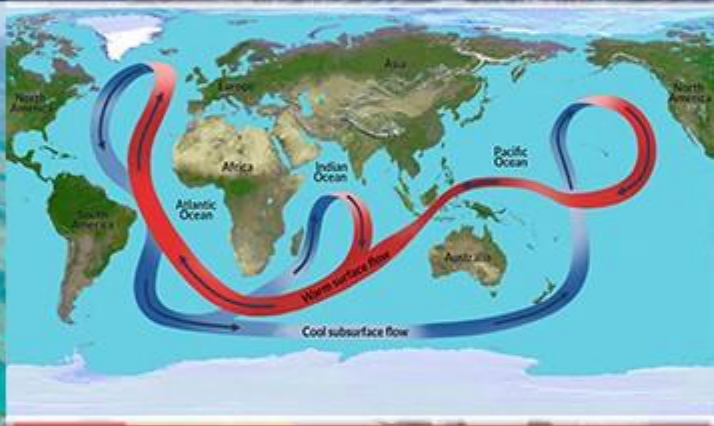
GEOGRAPHY

PHYSICAL GEOGRAPHY



OCEANOGRAPHY

MODULE - 5

- TYPES OF TIDES
- TIDAL BORE PHENOMENA

MOVEMENTS OF OCEAN WATER

TYPES OF TIDES

BASED ON FREQUENCY

- Semi-diurnal tide
- Diurnal tide
- Mixed tide

BASED ON THE SUN, MOON AND THE EARTH POSITIONS

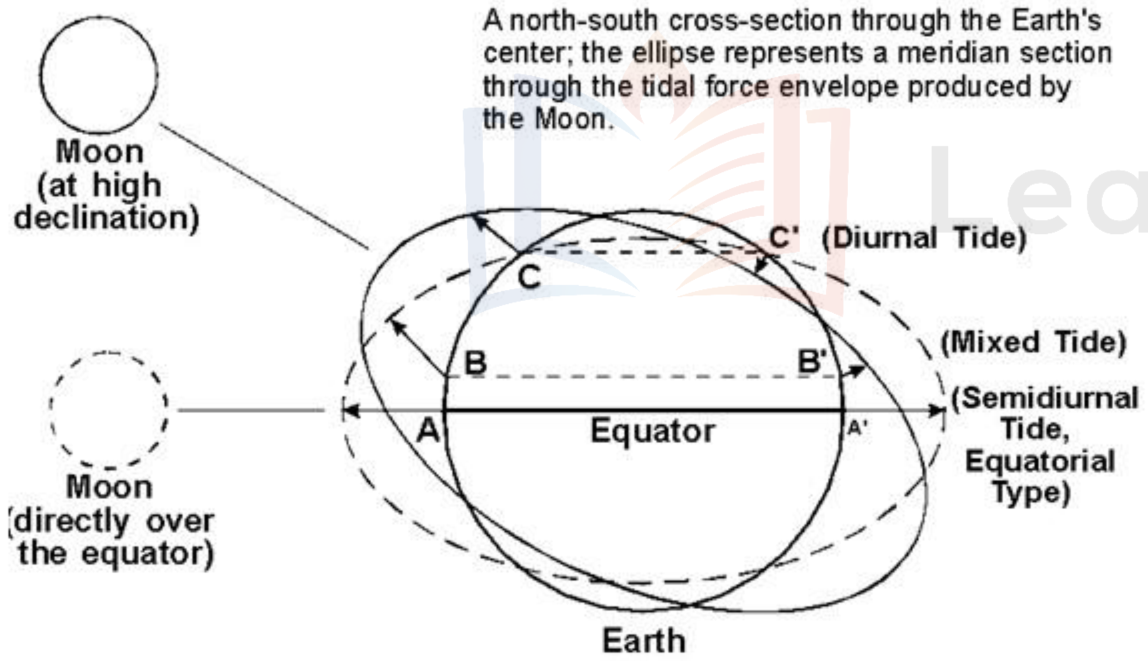
- Spring tides
- Neap tides

1. Tides vary in their frequency, direction and movement from place to place and also from time to time
2. Tides may be grouped into various types based on their frequency of occurrence in one day or 24 hours or based on the relative positions of the sun, moon and earth.

MOVEMENTS OF OCEAN WATER

TYPES OF TIDES

TIDES BASED ON FREQUENCY



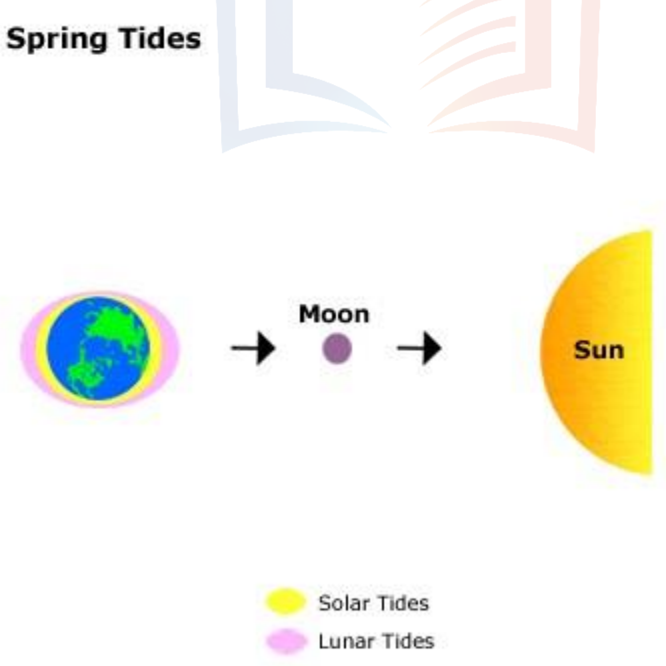
1. **Semi-diurnal tide :** The most common tidal pattern, featuring two high tides and two low tides each day. The successive high or low tides are approximately of the same height.
2. **Diurnal tide :** There is only one high tide and one low tide during each day. The successive high and low tides are approximately of the same height.
3. **Mixed tide :** Tides having variations in height are known as mixed tides. These tides generally occur along the west coast of North America and on many islands of the Pacific Ocean.

MOVEMENTS OF OCEAN WATER

TYPES OF TIDES

TIDES BASED ON THE SUN, MOON AND THE EARTH POSITIONS

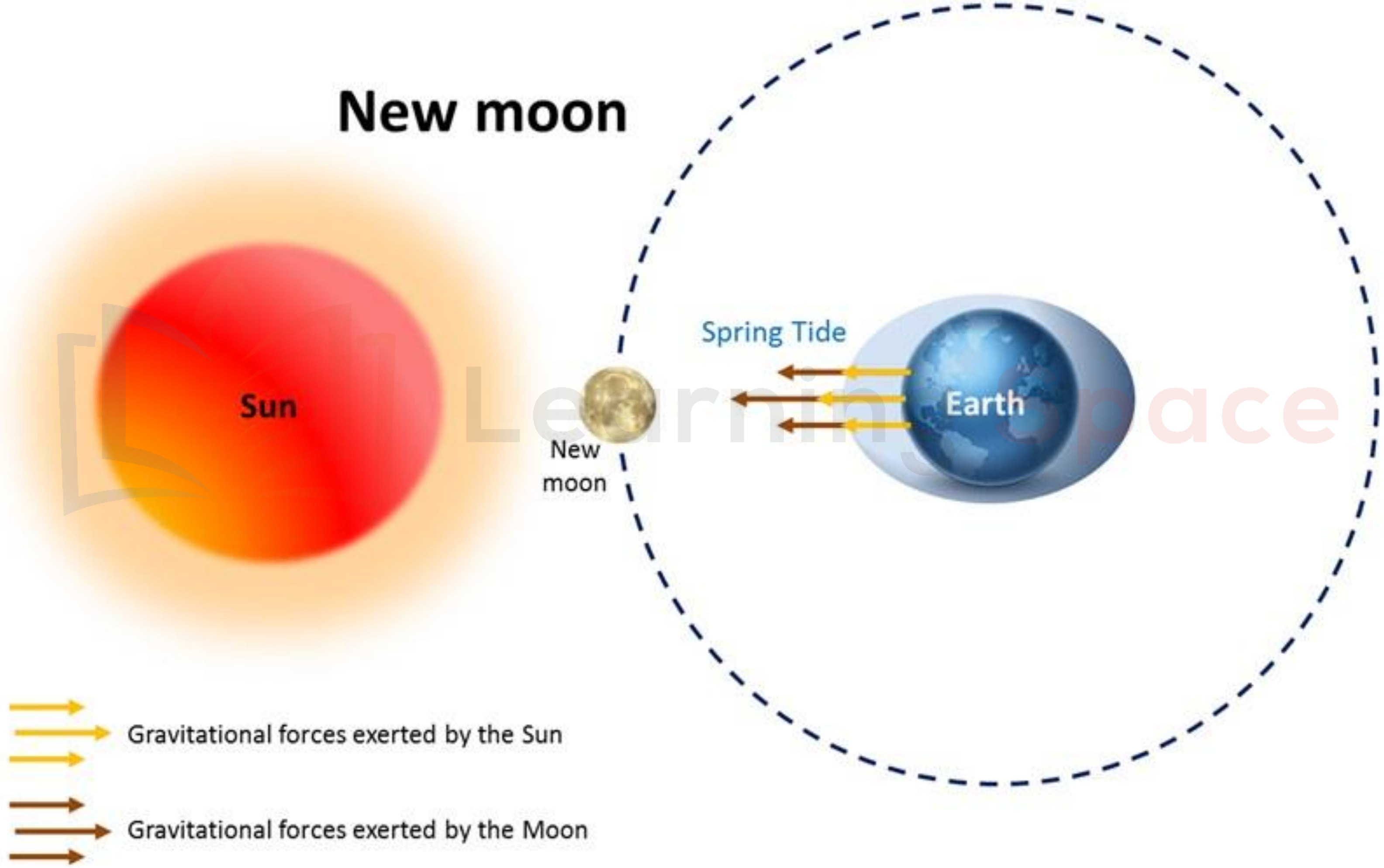
The height of rising water (high tide) varies appreciably depending upon the position of sun and moon with respect to the earth. Spring tides and neap tides come under this category.



1 Spring tides :

- The position of both the sun and the moon in relation to the earth has direct bearing on tide height.
- When the sun, the moon and the earth are in a straight line, the height of the tide will be higher.
- These are called spring tides and they occur twice a month, one on full moon period and another during new moon period.

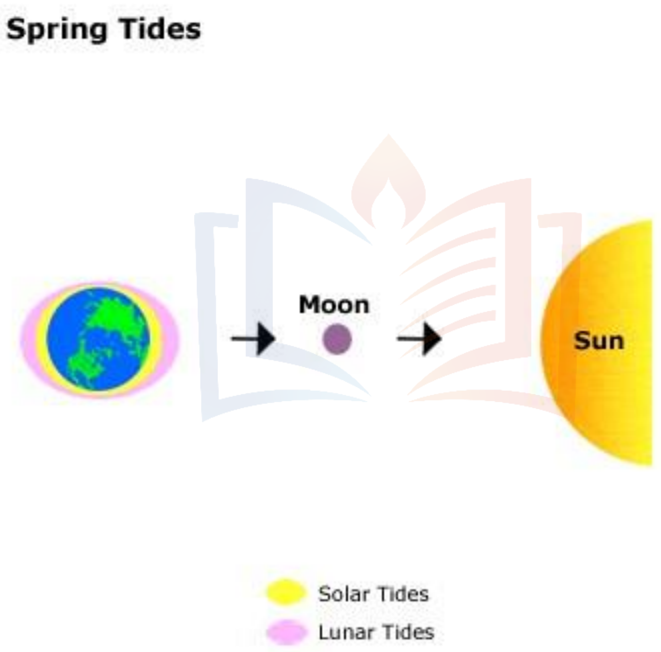
MOVEMENTS OF OCEAN WATER



MOVEMENTS OF OCEAN WATER

TYPES OF TIDES

TIDES BASED ON THE SUN, MOON AND THE EARTH POSITIONS



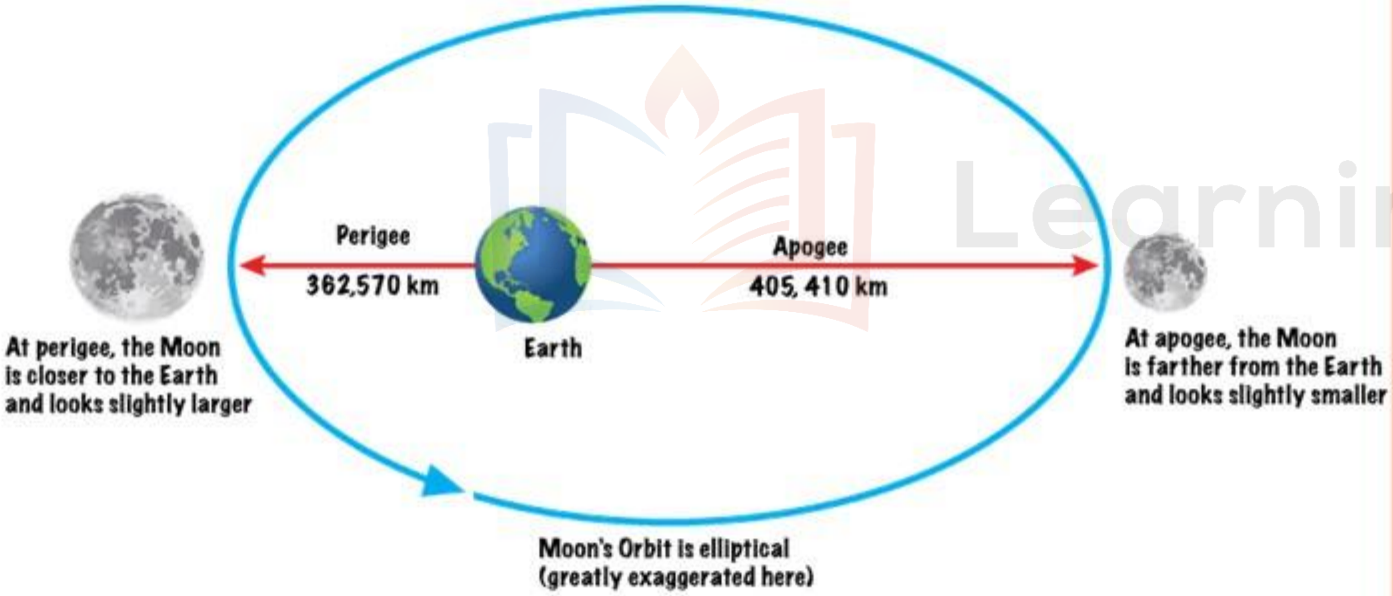
2 Neap tides :

- Normally, there is a seven day interval between the spring tides and neap tides. At this time the sun and moon are at right angles to each other and the forces of the sun and moon tend to counteract one another.
- The Moon's attraction, though more than twice as strong as the sun's, is diminished by the counteracting force of the sun's gravitational pull.

MOVEMENTS OF OCEAN WATER

TYPES OF TIDES

TIDES BASED ON THE SUN, MOON AND THE EARTH POSITIONS



3. Once in a month, when the moon's orbit is closest to the earth (perigee), unusually high and low tides occur.
4. During this time the tidal range is greater than normal.
5. Two weeks later, when the moon is farthest from earth *(apogee)*, the moon's gravitational force is limited and the tidal ranges are less than their average heights.

MOVEMENTS OF OCEAN WATER

TYPES OF TIDES

TIDES BASED ON THE SUN, MOON AND THE EARTH POSITIONS

6 When the earth is closest to the sun *(perihelion)*, around 3rd January each year, tidal ranges are also much greater, with unusually high and unusually low tides.

7 When the earth is farthest from the sun (aphelion), around 4th July each year, tidal ranges are much less than average.

8 The time between the high tide and low tide, when the water level is falling, is called the ebb.

9 The time between the low tide and high tide, when the tide is rising, is called the flow or flood.

MOVEMENTS OF OCEAN WATER

IMPORTANCE OF TIDES

1. Since tides are caused by the earth-moon-sun positions which are known accurately, the tides can be predicted well in advance.
2. This helps the navigators and fishermen plan their activities.
3. Tidal flows are of great importance in navigation.
4. Tidal heights are very important, especially for harbours near rivers.
5. Tides are also helpful in desilting the sediments and in removing polluted water from river estuaries.
6. Tides are used to generate electrical power (in Canada, France, Russia, and China).

MOVEMENTS OF OCEAN WATER

IMPORTANCE OF TIDES

1 NAVIGATION

1. Tidal heights are very important, especially harbours near rivers and within estuaries having shallow 'bars' at the entrance, which prevent ships and boats from entering into the harbour.
2. High tides help in navigation. They raise the water level close to the shores. This helps the ships to arrive at the harbour more easily.
3. Tides generally help in making some of the rivers navigable for ocean-going vessels. London and Calcutta have become important ports owing to the tidal nature of the mouths of the Thames and Hooghly respectively.

Learning Space

MOVEMENTS OF OCEAN WATER

IMPORTANCE OF TIDES

2 FISHING

1. The high tides also help in fishing.
2. Many more fish come closer to the shore during the high tide.
3. This enables fishermen to get a plentiful catch.

3 DESILTING

Tides are also helpful in desilting the sediments and in removing polluted water from river estuaries.

4 POWER

1. Tides are used to generate electrical power (in Canada, France, Russia, and China).
2. A 3 MW tidal power project was planned to be constructed at Durgaduani in Sunderbans of West Bengal.

MOVEMENTS OF OCEAN WATER

WHAT DO YOU MEAN BY TIDAL BORE?

1 Tides also occur in gulfs. The gulfs with wide fronts and narrow rears experience higher tides.

2 The in and out movement of water into a gulf through channels is called a tidal current.

3 When a tide enters the narrow and shallow estuary of a river, the front of the tidal wave appears to be vertical owing to the piling up of water of the river against the tidal wave and the friction of the river bed.

4 The steep-nosed tide crest looks like a vertical wall of-water rushing upstream and is known as a TIDAL BORE.

MOVEMENTS OF OCEAN WATER

WHAT DO YOU MEAN BY TIDAL BORE?

Tidal bore in Nova Scotia

Tidal bore in a bay in United Kingdom

MOVEMENTS OF OCEAN WATER

WHAT DO YOU MEAN BY TIDAL BORE?

Tidal bore in Hugli River

5. The favorable conditions for tidal bore include strength of the incoming tidal wave, depth of the channel and the river flow.
6. There are exceptions. The Amazon River is the largest river in the world. It empties into the Atlantic Ocean.
7. The mouth of the Amazon is not narrow, but the river still has a strong tidal bore. A tidal bore develops here because the the river is shallow in nature.

MOVEMENTS OF OCEAN WATER

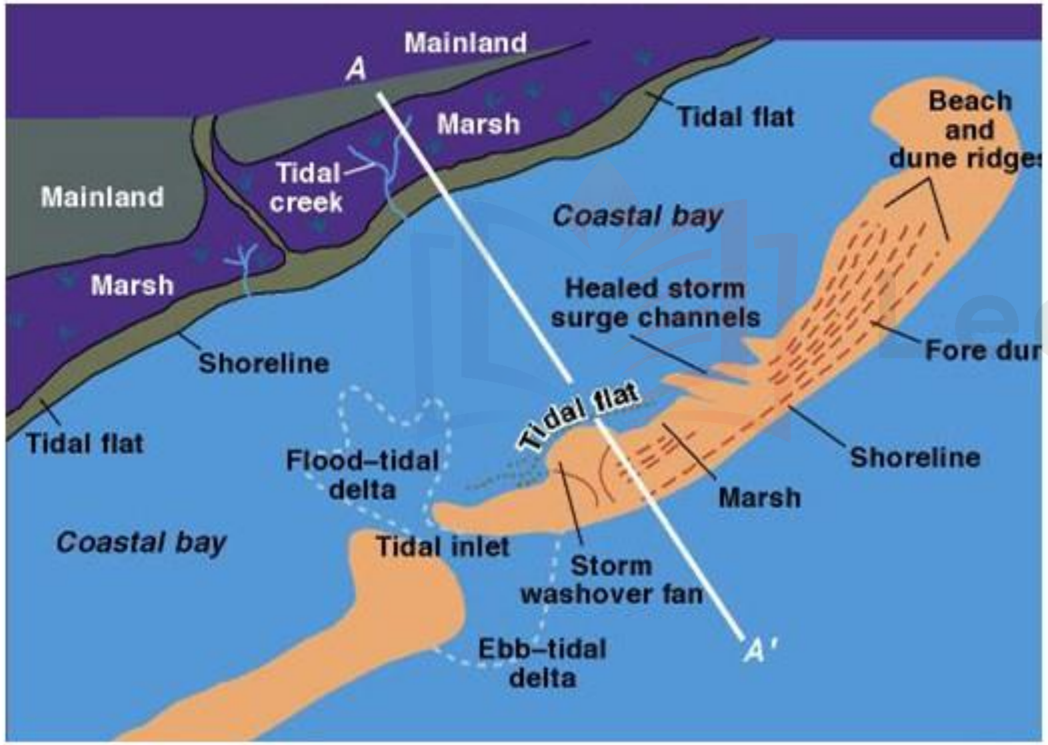
Tidal bore in Qiantang river

VIDEO CREDITS:
CCTV Video News Agency

8 In India, tidal bores are common in the Hooghly river. Most powerful tidal bores occur in Qiantang River in China.

MOVEMENTS OF OCEAN WATER

WHAT DO YOU MEAN BY TIDAL BORE?



9 The name 'bore' is because of the sound the tidal current makes when it travels through narrow channels.

10 Bores occur in relatively few locations worldwide, usually in areas with a large tidal range, typically more than 6 metres (20 ft) between high and low water.

11 A tidal bore takes place during the flood tide and never during the ebb tide

12 Tidal bores almost never occur during neap tides. Neap tides happen during quarter moons, when tides are weakest.

MOVEMENTS OF OCEAN WATER

IMPACT OF TIDAL BORE

1. Tides are stable and can be predicted. Tidal bores are less predictable and hence can be dangerous.
2. The tidal bores adversely affect the shipping and navigation in the estuarine zone.
3. Tidal bores of considerable magnitude can capsize boats and ships of considerable size.
4. Strong tidal bores disrupt fishing zones in estuaries and gulfs.
5. Tidal bores have an adverse impact on the ecology of the river mouth.
6. Animals slammed by the leading edge of a tidal wave can be found dead or dazed in the silty water.
6. That is why, carnivores and scavengers are common sights behind tidal bores.
7. In the Amazon, piranhas gobble up fish, crabs, and even birds left behind by the wave. Bears and eagles wade into the water hours after the wave passes to pick up fish along the banks.

WE EXPRESS OUR SINCERE THANKS FOR VIEWING THIS VIDEO

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